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1. What is an arithmetic sequence? 1 / 5 points

- ☐ A sequence where each term is the sum of all previous terms.
- ☐ A sequence where each term is a product of the previous terms.
- ☐ A sequence where each term is divided by a constant.
- ☒ A sequence where each term is obtained by adding a constant to the previous term.
- ☐ Nice work

Correct. In an arithmetic sequence, each term is obtained by adding a fixed constant to the previous term.

2. Which term represents the common difference in an arithmetic sequence? 1 / 5 points

- ☐ a
- ☐ n
- ☒ d
- ☐ r

☐ Nice work

Correct. d represents the common difference in an arithmetic sequence.

3. What is the first term of an arithmetic sequence called? 1 / 5 points

- ☐ r
- ☐ d
- ☒ a
- ☐ n

☐ Nice work

Correct. a represents the first term of the sequence.

4. How is the n th term of an arithmetic sequence represented? 1 / 5 points

- ☐ $a \cdot r^{n-1}$
- ☒ $a + (n - 1)d$
- ☐ $a + (n - 1)d$
- ☐ $a + nd$

☐ Nice work

Correct. This is the formula for the n th term of an arithmetic sequence.

5. What is a geometric sequence? 1 / 5 points

- ☐ A sequence where each term is divided by a constant.
- ☐ A sequence where each term is added by a constant.
- ☐ A sequence where each term is subtracted by a constant.
- ☒ A sequence where each term is multiplied by a constant.
- ☐ Nice work

Correct. In a geometric sequence, each term is obtained by multiplying the previous term by a fixed constant.

6. What does the common ratio in a geometric sequence represent? 1 / 5 points

- ☐ The sum of terms.
- ☐ The difference between terms.
- ☐ The product of terms.
- ☒ The ratio between consecutive terms.
- ☐ Nice work

Correct. The common ratio is the ratio between consecutive terms in a geometric sequence.

7. How is the n th term of a geometric sequence represented? 1 / 5 points

- ☒ $a \cdot r^{n-1}$
- ☐ $a \cdot (n - 1)$
- ☐ a/r^{n-1}
- ☐ $a \cdot (n - 1)d$

☐ Nice work

Correct. This is the formula for the n th term of a geometric sequence.

8. What happens to a geometric sequence if the common ratio r is greater than 1? 1 / 5 points

- ☐ It fluctuates between values.
- ☒ It diverges to infinity.
- ☐ It converges to a number.
- ☐ It remains constant.
- ☐ Nice work

Correct. The sequence diverges to infinity if the common ratio is greater than 1.

9. What is a Fibonacci sequence? 1 / 5 points

- ☐ A sequence where each term is the quotient of the two preceding terms.
- ☒ A sequence where each term is the sum of the two preceding terms.
- ☐ A sequence where each term is the difference of the two preceding terms.
- ☐ A sequence where each term is the product of the two preceding terms.
- ☐ Nice work

Correct. This is the definition of the Fibonacci sequence.

10. Find the 10th term of an arithmetic sequence where the first term is 5 and the common difference d is 3. 1 / 5 points

32

☐ Nice work

Step 1: Identify the values:

$a = 5$
 $d = 3$
 $n = 10$

Step 2: Apply the formula for the n th term:

$u_n = a + (n - 1)d$
 $u_{10} = 5 + (10 - 1) \cdot 3$
 $= 5 + 9 \cdot 3$
 $= 5 + 27$
 $= 32$

11. Determine the 5th term of a geometric sequence where the first term is 2 and the common ratio r is 3. 1 / 5 points

162

☐ Nice work

Step 1: Identify the values:

$a = 2$
 $r = 3$
 $n = 5$

Step 2: Apply the formula for the n th term:

$u_n = a \cdot r^{n-1}$
 $u_5 = 2 \cdot 3^{5-1}$
 $= 2 \cdot 3^4$
 $= 2 \cdot 81$
 $= 162$

12. If the 4th term of an arithmetic sequence is 28 and the common difference d is 5, find the first term. 1 / 5 points

5

☐ Nice work

Step 1: Identify the values:

$u_4 = 28$
 $d = 5$

Step 2: Apply the formula for the n th term:

$u_n = a + (n - 1)d$
 $u_4 = a + 3d$
 $28 = a + 3 \cdot 5$
 $28 = a + 15$
 $a = 28 - 15$
 $a = 5$

13. Calculate the sum of the first 8 terms of an arithmetic sequence with $a = 2$ and $d = 6$. 1 / 5 points

72

☐ Nice work

Step 1: Identify the values:

$a = 2$
 $d = 6$
 $n = 8$

Step 2: Apply the formula for the sum of the first n terms:

$S_n = \frac{n}{2} (2a + (n - 1)d)$
 $S_8 = \frac{8}{2} (2 \cdot 2 + 5 \cdot 6)$
 $= 4(4 + 30)$
 $= 4 \cdot 34$
 $= 72$

14. Find the sum of the first four terms of a geometric sequence where $a = 1$ and $r = 2$. 1 / 5 points

15

Feedback

Step 1: Identify the values:

$a = 1$
 $r = 2$
 $n = 4$

Step 2: Apply the formula for the sum of the first n terms:

$S_n = a \frac{r^n - 1}{r - 1}$
 $S_4 = 1 \cdot \frac{2^4 - 1}{2 - 1}$
 $= \frac{16 - 1}{1}$
 $= 15$

15. Determine the 10th term of a Fibonacci sequence starting with $u_1 = 0$ and $u_2 = 1$. 1 / 5 points

5

Feedback

Step 1: Identify the values and apply the Fibonacci sequence:

$u_1 = 0$
 $u_2 = 1$

Step 2: Calculate each subsequent term:

$u_3 = u_1 + u_2 = 1 + 0 = 1$
 $u_4 = u_2 + u_3 = 1 + 1 = 2$
 $u_5 = u_3 + u_4 = 2 + 1 = 3$
 $u_6 = u_4 + u_5 = 3 + 2 = 5$

16. For the sequence defined by $u_n = n^2 - n + 1$, find u_5 . 1 / 5 points

21

☐ Nice work

Step 1: Identify the formula and apply the given term:

$u_n = n^2 - n + 1$
 $n = 5$

Step 2: Calculate the specific term:

$u_5 = 5^2 - 5 + 1$
 $= 25 - 5 + 1$
 $= 21$

17. If $u_n = 2^n + 3$, calculate u_5 . 1 / 5 points

17

☐ Nice work

Step 1: Identify the formula and apply the given term:

$u_n = 2^n + 3$
 $n = 5$

Step 2: Calculate the specific term:

$u_5 = 2^5 + 3$
 $= 16 + 3$
 $= 19$

18. Find the limit of the sequence $u_n = \frac{1}{n}$ as n approaches infinity. 1 / 5 points

0

☐ Nice work

19. Given a sequence where $u_{n+1} = 3u_n + 2$ and $u_1 = 1$, find u_5 . 1 / 5 points

17

☐ Nice work

Step 1: Identify the recursive formula:

$u_1 = 1$
 $u_{n+1} = 3u_n + 2$

Step 2: Calculate the next terms:

$u_2 = 3u_1 + 2$
 $= 3 \cdot 1 + 2$
 $= 5$
 $u_3 = 3u_2 + 2$
 $= 3 \cdot 5 + 2$
 $= 17$